All in Books

Personal Accounting App — Android & Web

Product Requirements Document • v3.0 • March 2026 • FINAL

Currency	Platform	iOS	Status
INR (₹) Only	**Android + Web**	**Not Supported**	**FINAL**

Features removed vs v2: OCR auto-fill • Multi-book support • Cash book / Bank book module

1. Executive Summary & Vision

All in Books is a free, offline-first Progressive Web App (PWA) for personal and small-business accounting in India. It runs on Android Chrome (installable PWA) and any desktop web browser — one codebase, no backend server, no app store. Currency is fixed to INR (₹). The user always owns their data, stored on-device and backed up to their own Google Drive.

> *"All in Books should feel like the best ledger book you have ever used — except it lives on your Android phone, never loses a page, and hands you a professional report whenever you need one."*

1.1 Resolved Design Decisions

ID	Topic	Decision
D-001	**App Name**	All in Books
D-002	**Platform**	Android Chrome (PWA) + Desktop Web. iOS excluded — all phases.
D-003	**Currency**	INR (₹) only. No multi-currency.
D-004	**Invoice Template**	One single professional template. No selector.
D-005	**Historical Import**	5-sheet .xlsx template. User fills → uploads once.
D-006	**Attachment Size**	Per-file cap 50 MB. App blocks and shows error if exceeded.
D-007	**Recurring Entries**	User must confirm each due entry via modal. No silent auto-posting.
D-008	**OCR**	REMOVED from all phases.
D-009	**Multi-book Support**	REMOVED. Single book only. bookId="main" hardcoded.
D-010	**Cash Book / Bank Book**	REMOVED. Out of scope for all phases.

2. User Personas

Persona 1 — Arjun (Kirana Shop Owner, age 34)

Situation: Runs grocery store. 8–15 vendors, 40–60 customers/day. Physical bahi-khata. WhatsApp-comfortable.
Pain points: Loses track of who owes money; cannot calculate monthly profit; no digital record for CA.
Goals: Record transactions in <30 sec; outstanding balances at a glance; WhatsApp reminders; Excel export for CA.

Persona 2 — Priya (Freelance Designer, age 27)

Situation: Issues invoices, receives partial payments, tracks expenses. Files ITR independently.
Pain points: Invoice disputes; no paid/unpaid tracking; receipts lost before tax season.
Goals: Professional invoices; attach expense receipts; clean GST summary at year-end.

Persona 3 — Meena (Tiffin Service Owner, age 42)

Situation: Manages household budget and a tiffin delivery business. No accounting background.
Pain points: Cannot separate personal/business expenses; does not know if tiffin is profitable; forgets to collect.
Goals: Simple daily entry; see who owes money; monthly profit view.

3. Feature Registry

P0 = MVP must-have (Phase 1). P1 = pre-public-release (Phase 1–2). P2 = post-launch (Phase 2+). OCR, multi-book, and cash book have been permanently removed.

ID	Feature	Category	Pri	Phase
F-001	**Journal Entry — Sale**	Core	P0	Phase 1
F-002	**Journal Entry — Purchase**	Core	P0	Phase 1
F-003	**Journal Entry — Expense**	Core	P0	Phase 1
F-004	**Journal Entry — Receipt**	Core	P0	Phase 1
F-005	**Journal Entry — Payment**	Core	P0	Phase 1
F-006	**GST fields on all entries (rate, base, gst amount auto-calc)**	Core	P0	Phase 1
F-007	**Master Ledger View**	Core	P0	Phase 1
F-008	**Individual Party Ledger**	Core	P0	Phase 1
F-009	**Party Management (name, phone, address, GSTIN, opening balance)**	Core	P0	Phase 1
F-010	**Party Statement + PDF Share via Android share sheet**	Core	P0	Phase 1
F-011	**Invoice Creation (line items, GST, discount, auto-number)**	Core	P0	Phase 1
F-012	**Invoice PDF — single professional template, fully on-device**	Core	P0	Phase 1
F-013	**WhatsApp Share via Android native share sheet (free, no API cost)**	Core	P0	Phase 1
F-014	**Photo Bill Attachment (50 MB max, compress to <400 KB)**	Core	P0	Phase 1
F-015	**Offline Mode — full PWA functionality, zero network dependency**	Core	P0	Phase 1
F-016	**Excel Export — 7-sheet workbook via SheetJS**	Core	P0	Phase 1
F-017	**Google Drive Auto-Backup (weekly, rolling 4-week retention)**	Core	P0	Phase 1
F-018	**P&L Statement — period picker + daily bar chart**	Reports	P0	Phase 1
F-019	**GST / Tax Summary by rate slab (5 / 12 / 18 / 28 %)**	Reports	P0	Phase 1
F-020	**Historical Import — 5-sheet Excel template upload**	Import	P0	Phase 1
F-021	**Import Pre-import Validation Review Screen**	Import	P0	Phase 1
F-022	**Import Post-import Summary + 24-hour Undo**	Import	P0	Phase 1
F-023	Simplified Balance Sheet / Net Position view	Reports	P1	Phase 1–2
F-024	Category Ledger View	Core	P1	Phase 1–2
F-025	Invoice Status Tracking (Draft / Sent / Partial / Paid)	Core	P1	Phase 1–2

ID	Feature	Category	Pri	Phase
F-026	Recurring Entries with user-confirmation modal	Optional	P2	Phase 2
F-027	**PWA Install — Android Chrome home screen icon**	Core	P0	Phase 1
F-028	**Service Worker + Offline Asset Cache**	Core	P0	Phase 1
F-029	**3-Step Onboarding Flow**	Core	P0	Phase 1

4. User Stories & Acceptance Criteria

4.1 Journal Entries

US-001 *As a shop owner, I want to quickly record a sale transaction, so that I have an accurate searchable record of all income.*

Acceptance Criteria:

1. Entry form opens within 1 tap via FAB from any screen
2. Required fields: type and amount only — all others optional
3. Entry type defaults to last-used; date defaults to today (editable)
4. Transaction saved to IndexedDB immediately on tapping Save — no server call
5. Green toast 'Saved ✓' appears for 2 seconds and auto-dismisses

US-002 *As a GST-registered business owner, I want to record the GST component separately from the base amount, so that I can calculate my net GST liability at any time.*

Acceptance Criteria:

6. GST toggle defaults Off; toggling On shows rate chips: 0%, 5%, 12%, 18%, 28%
7. User enters total (GST-inclusive); base and GST auto-calculated live via calcFromTotal()
8. Both baseAmount and gstAmount stored as separate numeric fields in IndexedDB
9. If GST off, gstRate and gstAmount stored as null — not zero
10. GST breakdown visible in entry detail, ledger export, and GST Summary report

US-003 *As a user, I want to attach a photo of a bill to any entry, so that I have a permanent digital copy without keeping paper.*

Acceptance Criteria:

11. Camera (Android) or gallery file selection available on every entry form
12. Files over 50 MB blocked with toast error: 'File too large. Max size is 50 MB.'
13. Valid attachments compressed to under 400 KB before IndexedDB storage
14. Thumbnail shown in entry list; tapping opens full-res lightbox via Modal

4.2 Historical Data Import

US-004 *As a user migrating from paper or Excel, I want to upload a structured Excel file to import all historical data, so that I do not lose years of records when switching to All in Books.*

Acceptance Criteria:

15. Import accessible from Settings → Data & Backup → Import Historical Data
16. App provides downloadable import template (.xlsx) — always available offline
17. Template has 5 sheets: Transactions, Party_Balances, Parties, GST_History, README
18. Pre-import review screen: green (valid), red (error) rows with per-cell error messages
19. User may proceed with valid rows only, or cancel entirely
20. Post-import summary: Imported N, Skipped (duplicates) N, Errors N
21. 'Undo Import' button available for 24 hours after import completes

4.3 Parties & Ledger

US-005 *As a user, I want to see a complete statement of all transactions with a party, so that I always know exactly how much I owe or am owed.*

Acceptance Criteria:

22. Statement: date, description, DR, CR, running balance starting from opening balance
23. Outstanding balance prominent at top — green (to receive) or red (to pay)
24. Share button generates PDF on-device, opens Android native share sheet
25. Statement filterable by date range; imported entries show "Imported" badge

4.4 Invoice Creation

US-006 *As a freelancer, I want to create a professional invoice and share it via WhatsApp, so that my client receives a clear payment request and I have a record.*

Acceptance Criteria:

26. Fields: party, line items (name/qty/rate), GST toggle+rate, discount, due date, notes
27. Invoice number auto-increments: AIB-0001, AIB-0002 … (prefix editable in Settings)
28. PDF generated entirely on-device via jsPDF + html2canvas — no server call
29. Single professional template: logo top-left, party top-right, line-item table, bold total
30. 'Share via WhatsApp' opens Android native share sheet with PDF attached
31. 'Record as Sale' creates a linked journal entry and marks invoice as Sent

4.5 Offline Mode

US-007 *As a user in a low-connectivity area, I want to use all app features without internet, so that I am never blocked from recording transactions.*

Acceptance Criteria:

32. App loads fully from service worker cache with zero network
33. All reads and writes go to IndexedDB — no network calls on any core operation
34. Offline dot visible in header when navigator.onLine is false
35. Google Drive sync fires automatically in background when connectivity restores
36. No loading spinners or degraded UI due to offline status

4.6 Export & Backup

US-008 *As a user, I want to export all data to Excel and back it up to Google Drive weekly, so that I never lose data and my accountant can read it.*

Acceptance Criteria:

37. Excel export: 7 sheets — Transactions, Parties, Invoices, Categories, GST_Summary, Monthly_Summary, Import_Log
38. All amounts stored as Excel numbers; all dates as Excel date values (not text strings)
39. File named: AllInBooks_Export_[MMM-YYYY].xlsx
40. Auto-backup: every Sunday on next app open; rolling 4 weekly + 3 monthly files retained
41. 'Manual 'Backup Now' button available; last backup timestamp shown in Settings

4.7 Reports

US-009 *As a user, I want to see P&L and GST summary for any period, so that I understand whether my business made money and what I owe in tax.*

Acceptance Criteria:

42. Period picker: This Week, This Month, Last Month, Custom Range
43. P&L: Total Revenue, Total Expenses, Net Profit/Loss as large numbers; daily bar chart below
44. GST Summary: Output GST, Input GST, Net Payable — broken down by rate slab
45. Both reports export to PDF in one tap

4.8 Recurring Entries (Phase 2)

US-010 *As a user with fixed monthly expenses, I want to set up a recurring entry for rent or salary, so that I do not forget predictable transactions.*

Acceptance Criteria:

46. User sets: type, amount, party (optional), frequency (weekly/monthly), start date
47. On due date, on next app open, a confirmation modal appears: 'Record it now?'
48. User taps 'Record' (saves entry) or 'Skip This Time' (no entry, date advances)
49. No entry is ever created without explicit user confirmation — no silent auto-posting

5. Import Template Specification

Sheet 1: Transactions

Columns: date (YYYY-MM-DD, required), type (sale|purchase|expense|receipt|payment, required), amount (positive number, required), baseAmount (optional), gstRate (0/5/12/18/28/blank, optional), gstAmount (optional), partyName (optional), category (optional), note (max 500 chars), invoiceNumber (optional).

Sheet 2: Party_Balances

Columns: partyName (required), balanceDate (YYYY-MM-DD, required), balanceAmount (positive, required), balanceType (toReceive|toPay, required), note (optional).

Sheet 3: Parties

Columns: name (required), phone (optional), address (optional), gstin (optional).

Sheet 4: GST_History

Columns: period (e.g. Q3 FY2025, required), outputGST (number, required), inputGST (number, required), gstRate (optional), note (optional).

Sheet 5: README

Non-data sheet. Plain English step-by-step guide, field instructions, example data. Row 1=headers, Row 2=format hints, Row 3=example in all data sheets.

Validation Rules

Rule	Condition	Behaviour
V-001	Required field blank (date/type/amount)	Row: error. Not imported.
V-002	Invalid enum (type, balanceType)	Row: error. Allowed values listed in message.
V-003	Amount is zero or negative	Row: error.
V-004	GST mismatch — base+gst ≠ total (±₹1 tolerance)	Row: warning. User can override.
V-005	Unknown party name	New party auto-created. Not flagged as error.
V-006	Duplicate transaction (same date+type+amount+party)	Row: skipped. Counted separately.
V-007	File exceeds 50 MB	Rejected before parsing.
V-008	File is not .xlsx	Rejected immediately.
V-009	Required sheet missing (Transactions)	File-level error. No rows processed.

6. Data Model — IndexedDB via Dexie.js

Currency is always INR — no currency field. bookId is always "main" (single book, hardcoded). All tables below.

Table: `transactions`

Dexie Index String	All Stored Fields
'++id, bookId, type, date, partyId, categoryId, isImported, importBatchId'	id, bookId, type, date, partyId, partyName, categoryId, amount, baseAmount, gstRate, gstAmount, note, photoKey, invoiceId, isImported, importBatchId, importedAt, createdAt, updatedAt

Table: `parties`

Dexie Index String	All Stored Fields
'++id, bookId, name'	id, bookId, name, phone, address, gstin, openingBalance, openingBalanceDate, createdAt

Table: `invoices`

Dexie Index String	All Stored Fields
'++id, bookId, partyId, status, invoiceNumber'	id, bookId, invoiceNumber, partyId, lineItems (JSON), subtotal, gstRate, gstAmount, discount, total, issueDate, dueDate, status, notes, linkedTransactionId

Table: `categories`

Dexie Index String	All Stored Fields
'++id, bookId, type'	id, bookId, name, type, color, isDefault

Table: `importBatches`

Dexie Index String	All Stored Fields
'++id, bookId, importedAt'	id, bookId, importedAt, rowsImported, rowsSkipped, rowsErrored, undoneAt

Table: `recurringEntries`

Dexie Index String	All Stored Fields
'++id, bookId, nextDueDate, isActive'	id, bookId, type, amount, partyId, categoryId, frequency, nextDueDate, isActive, note

Table: `attachments`

Dexie Index String	All Stored Fields
'++id, transactionId'	id, transactionId, blob (Uint8Array), mimeType, originalName

7. Tech Stack & Architecture

Layer	Tool	Rationale
Framework	**React 18 + Vite**	Fast builds; AI-coder friendly; one codebase for web + PWA
Styling	**Tailwind CSS**	Utility-first; mobile-first; no separate CSS files
Local DB	**Dexie.js (IndexedDB)**	Offline-first; 22 KB; native to Android Chrome; no setup
State	**Zustand**	Zero boilerplate; async-friendly; per-store subscriptions
Excel I/O	**SheetJS (xlsx CE)**	Reads import template + writes export; free tier covers all needs
PDF	**jsPDF + html2canvas**	On-device invoice PDF; no server required; captures React component
Cloud Backup	**Google Drive REST API + OAuth 2.0 PKCE**	User owns their data; free; no backend server needed
PWA	**vite-plugin-pwa**	Service worker + web manifest auto-generated from vite.config.js
Hosting	**Vercel (free tier)**	Zero-config CI/CD from GitHub; SPA routing via vercel.json
Charts	**Recharts**	Declarative React charts; responsive; works with Tailwind
Icons	**Lucide React**	Clean, tree-shakeable; consistent with Tailwind aesthetic
Native (Phase 2)	**Capacitor**	Same PWA codebase → Play Store APK with zero rewrite

7.1 Data Flow — Local-First

User Action → Dexie.js (IndexedDB, instant, offline) → UI re-renders → [scheduled] SheetJS Export → Google Drive Backup. No user-facing operation ever waits on a network response.

7.2 PWA Caching Strategy

- App assets (JS / CSS / fonts / icons): Cache-First — always served from service worker cache
- Google Drive API calls: Network-First with 10-second timeout — falls back gracefully
- Initial JS bundle target: < 500 KB transferred (Vite bundle analyser)
- Android Chrome: PWA install prompt captured via beforeinstallprompt event

8. Non-Functional Requirements

Metric	Target	How Measured
App cold start (service worker cached)	**< 2 seconds**	Lighthouse TTI, mid-range Android
Entry save to IndexedDB	**< 100 ms**	Perceived instant; no spinner
Excel export — 1,000 transactions	**< 5 seconds**	Chrome DevTools timing
Invoice PDF generation	**< 3 seconds**	Timed on-device render
Import of 500 template rows	**< 8 seconds**	Validation + DB write combined
Lighthouse PWA score	**≥ 90**	Post service worker activation
Initial JS bundle	**< 500 KB transferred**	Vite bundle analyser

Out of Scope — All Phases

- iOS Safari support
- OCR / auto-fill from photos (removed)
- Multi-book support (removed)
- Cash Book / Bank Book module (removed)
- Multi-user or collaborative access
- Double-entry bookkeeping
- Bank statement auto-parsing
- Inventory or stock tracking
- Multi-currency